

GR-ProtipaExams: A Multimodal Benchmark for Evaluating LLM Reasoning in Greek Entrance Examinations

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Abstract

The rapid advancement of Large Language Models (LLMs) imposes a thorough evaluation of their linguistic and analytical capabilities as well as constraints, particularly for low-resource languages like Greek. To address the limited availability of comprehensive and multimodal benchmarks in this domain, we introduce **GR-ProtipaExams**, a dataset consisting of questions from entrance exams for Greek Model and Experimental schools. This benchmark is employed to assess the performance of two LLMs, the Greek **KriKri-DPO-Context** and **Gemma-3-27B-IT**, across four subjects (Modern Greek, Mathematics, Physics, and Religious Studies) and diverse task formats (closed and open-ended), including multimodal elements. Our findings indicate that Gemma consistently outperforms KriKri, exhibiting a systematic performance gap of approximately 20% across most metrics, a gap largely attributed to the difference in model scale and architecture. Furthermore, multimodality presents a significant challenge; introducing visual context leads to a notable performance decline for both models, with their generation capabilities converging in open-ended multimodal tasks. Finally, regarding the distinction between question types, results demonstrate that models exhibit superior consistency and performance in closed-ended tasks. In contrast, performance drops significantly in open-ended generation tasks, particularly for KriKri model, which struggles to maintain structural consistency in text generation.

1 Introduction

Evaluation tasks are crucial tools in the field of natural language processing (NLP) to keep track of the progress in machine learning and communicate the potential issues, biases, or needs that may arise from these evaluations (Biderman et al., 2024). To enable this process, benchmarks are deployed as reference points for Large Language Models (LLMs) to assess their performance on an equal footing for comparison purposes. These mainly

consist of one or multiple datasets along with relevant metrics and a standardised methodology to evaluate their performance (Ruder, 2021).

Well-known benchmarks, such as General Language Understanding Evaluation (GLUE) (Wang et al., 2018) and Cross-lingual TRansfer Evaluation of Multilingual Encoders (XTREME) (Hu et al., 2020), are collections of tasks used for training, evaluation, and analysis of monolingual and/or multilingual language models in a series of NLU and NLP tasks. These benchmarks serve as an objective lens to observe each model’s suitability for a specific task as well as the field’s progression through remarkable findings.

However, issues regarding the transparency and reproducibility of LLM evaluations conducted by independent researchers have been noted. Specifically, concerns related to data contamination, where models may have seen test data during training, pose a major challenge to the validity of results (Sainz et al., 2023; Zhou et al., 2023). This highlights the need for a unified framework for result reproduction and novel evaluations on any LLM using already supported benchmarks (Biderman et al., 2024; Siddiq et al., 2025).

To this end, Gao et al. Gao et al. (2021) built the Language Model Evaluation Harness -known as “lm-eval”- which serves as an open-source research library for LLM evaluation. This infrastructure is publicly available, offering a unified framework to test generative language models on a large number of different evaluation tasks and compare the results across models in a standardised way.

Building upon the need for standardised assessment in non-English contexts, this project involves the creation and evaluation of a Greek dataset, called “GR-ProtipaExams”, which includes questions and answers from exams related to students’ admission to model and experimental schools in Greece, using the lm-eval framework ¹. This

¹The official Hugging Face dataset URL will be made publicly available upon acceptance.

dataset, as published online by the Greek Government from 2013 to 2025, includes topics related to the subjects of Greek Language, Math, Physics, and Religious Studies. Moreover, it covers levels of secondary education; specifically, the topics are divided into middle school (Gymnasium) and high school (Lyceum) entrance exams. By leveraging the benefits of the Im-eval infrastructure, we compare and evaluate the answers given by Krikripo-context and Gemma-3-27b-it LLMs to the GR-ProtipaExams benchmark in order to address the following research questions:

- **RQ1:** How does the performance of a specialized Greek model compare to a general-purpose LLM across the four subjects—Mathematics, Physics, Greek Language, and Religious Studies—in terms of reasoning, linguistic, and factual capabilities?
- **RQ2:** To what extent does the inclusion of textual descriptions for visual elements affect the language models’ accuracy compared to purely text-based questions?
- **RQ3:** How does performance vary across different exercise formats, ranging from closed-form questions to open-ended tasks?

2 Related Work

Since the release date of BERT (Devlin et al., 2019), a plethora of language models have emerged, ranging from large-scale models such as GPT-2 (Radford et al., 2019), GPT-3 (Brown et al., 2020), ChatGPT and Gemini (Team et al., 2025), to smaller-scale ones, such as Gemma (Gemma Team et al., 2025), GPT-Neo (Black et al., 2022), DistilBERT (Sanh et al., 2020) or T5-Small (Raffel et al., 2023). The emergence of these models highlights the need to systematically explore their capabilities in natural language understanding, processing, and generation, as well as their limitations.

Consequently, a series of notable benchmarks have been developed for LLM evaluation, focusing on various task types (e.g. open-ended, closed questions, image recognition) derived from academic and subject-specific fields (such as mathematics, science, languages, physics) or everyday tasks demanding reasoning and common sense. One of the earliest multimodal question-answering datasets is DAQUAR (Malinowski and Fritz, 2015), which involves image recognition and processing to answer simple questions about provided images correctly.

Following this, a number of prominent datasets have emerged assisting the exploration of LLMs capabilities and constraints. Specifically, the TQA dataset was first introduced in 2017 (Kembhavi et al., 2017), involving the task of multimodal machine comprehension given the relevant textual context (images, diagrams). The subjects (Life Science, Earth Science and Physical Science) were derived from middle school science textbooks, and the dataset consists of 1,076 lessons in total.

In the field of Natural Language Inference (NLI), the Hellaswag dataset (Zellers et al., 2019), introduced in 2019, addresses the task of common sense reasoning in everyday applications, shedding light on the inner mechanisms of deep pre-trained models as to how they complete a sentence given a description. By applying adversarial filtering to the BERT-Large model, the authors compared the answers across five models (OpenAI GPT, Bert-Base, ESIM+ELMo, LSTM sentence encoder, Fast-Text) with human performance in terms of ending generation accuracy; findings reveal that human performance remained superior, with BERT-Large achieving the highest score among the models.

Regarding Natural Language Understanding (NLU), the MMLU dataset, proposed by Hendrycks et al. (2021), introduced a new task for measuring the model’s multitask accuracy by evaluating them in zero-shot and few-shot settings. It involves multiple-choice questions from 57 different subjects across STEM, the humanities, and the social sciences, ranging from elementary to professional difficulty levels. In addition, the ScienceQA dataset (Lu et al., 2022) introduces approximately 21k multiple-choice questions derived from various science topics, providing relevant lectures and explanations as supplementary material to enhance model performance via CoT (Chain of Thought) prompting. Their findings indicate that few-shot language models benefit from these explanations, improving performance even with a smaller volume of data.

While a plethora of question-answering multilingual benchmarks like Belebele (Bandarkar et al., 2024) and XNLI (Conneau et al., 2018) exist, they are often limited in terms of scope, question format, and multimodal resources, especially for the Greek language. Existing Greek benchmarks mainly focus on machine translation between Ancient and Modern Greek (Anonymous, n.d.a), sentence-ending tasks such as the Greek version of HellaSwag (Anonymous, n.d.b), dialect

identification (Chatzikyriakidis et al., 2025), and Sign Language Translation (SLT) (Voskou et al., 2023). Most existing datasets cover a wide range of elementary school subjects, focusing on mathematics, where language models have already shown a strong performance (Clark et al., 2018; Mihaylov et al., 2018), as well as reading comprehension, such as the RACE (Lai et al., 2017) and MCTest (Richardson et al., 2013) datasets.

To address this gap, we introduce the “GR-ProtipaExams” dataset, which includes 1,646 entries covering the school subjects of Greek Language, Mathematics, Physics, and Religious Studies from middle and high school entrance exams. It covers a wide range of question types, open-ended and closed-form questions (multiple-choice, true/false, fill-in-the-gaps, and matching), which can be multimodal (i.e., containing images) or text-only. In addition, we provide input context as supplementary material to assist the model in understanding the provided texts and images, where needed to answer the question correctly.

By integrating GR-ProtipaExams into the Im-evaluation-harness infrastructure, we ensure that our evaluation process is transparent and easily reproducible by the broader research community, addressing the limitations noted in previous evaluation attempts (Gao et al., 2021).

3 Methodology

3.1 Data Collection and Processing

We gathered the corpus of the publicly available exams of the model and experimental school topics and organized them according to subject (Greek Language, Math, Physics, Religious Studies) and year (from 2013 to 2025). To ensure consistency, raw files were renamed using standardized conventions, which were subsequently used for the generation of unique IDs for each question-answer pair.

Each exam entry was processed and converted into a structured format. Specifically, questions were parsed into JSON files, while the corresponding solutions were extracted into Markdown files. The resulting dataset, named “GR-ProtipaExams”, preserves essential information for LLM training and evaluation through the following keys: id (unique identifier), question (task description), input (passages or context), choices (candidate answers for closed tasks), answer (the ground truth), image (path and metadata for multimodal entries),

and mark (grading score).

3.2 Dataset Statistics and Taxonomy

The “GR-ProtipaExams” dataset comprises a total of 1,646 entries. We classified these entries into two primary categories based on the required response type:

- Closed-Ended Questions (1,368 items): This category includes Multiple-Choice, True/False, Fill-in-the-gaps, and Matching tasks.
- Open-Ended Questions (278 items): This category consists of Development/Open questions and Fill-in-the-gaps tasks without provided choices.

The distribution varies across subjects: Modern Greek includes all task types, Mathematics and Physics contain both open and closed questions, while Religious Studies consists solely of closed (multiple-choice) items. Furthermore, the dataset supports multi-modality, indicating entries that require visual context (diagrams, geometric shapes, maps) for their resolution, totalling up to 270 multimodal QAs. Regarding this, two additional columns are provided in the dataset, called image_description and image_transcription, aiding the LLMs to visually process the provided image as well as any textual information.

3.3 Evaluation Setup

For the evaluation process, we utilized Im-eval-harness framework. We assessed the performance of two LLMs:

1. Gemma3-27b-it: A state-of-the-art general-purpose model.
2. Krikri-dpo-context: A model fine-tuned specifically for the Greek language.

For accuracy purposes, we designed two different task configurations within the evaluation harness:

- Closed-Task: Utilizes an evaluation prompt that instructs the model to output a specific index (e.g., A, B, 0, 1) corresponding to the correct choice. Performance is measured using Exact Match accuracy.

Subject	Total Qs	Accuracy (%)		ChrF	
		Gemma-3-27B	KriKri-DPO	Gemma-3-27B	KriKri-DPO
Greek Language	833	71.92	46.06	40.48	21.03
Mathematics	714	44.32	27.93	26.28	24.26
Physics	9	–	–	24.02	19.48
Religious Studies	90	77.78	62.22	–	–

Table 1: Comparison results between LLMs across the four subjects in zero-shot setup (RQ1).

- Open-Task: Deploys a generative prompt that guides the model to provide a full-text response or fill in the missing information. Performance is assessed using textual similarity metrics (e.g., BLEU, ChrF) against the ground truth.

3.4 Data Availability Limitations

During the data collection phase, certain inconsistencies were encountered with the source exam files; all files from the year 2015 were unavailable or corrupted, resulting in a gap in the dataset. Additionally, no solutions were provided for the Greek language subject for high schools in 2014, nor for the subjects of Greek and Math in 2018. Finally, essay/writing components of the Greek Language exams were excluded from the main dataset as no official answers are provided. For this reason, we compiled a separate file with the available writing prompts and grading guidelines for potential future qualitative analysis.

4 Experimental Results and Analysis

We present the evaluation results of the Gemma3-27b-it and Krikri-dpo-context models on the “GR-ProtipaExams” benchmark. The analysis is structured around the three research questions defined in the introduction, aiming to assess the models’ capabilities across different subjects, modalities, and reasoning tasks. For closed-ended questions, performance is reported in terms of Accuracy (Exact Match) against the ground truth, while for open-ended tasks, we employ textual similarity metrics (ChrF) to quantify overlaps between the generated output and the correct answer.

4.1 Comparative Performance Across Subjects

As depicted in Table 1, Gemma-3-27B outperforms KriKri-DPO-Context across all four subjects in both closed (Accuracy) and open-ended (ChrF)

tasks. This finding is expected, given Gemma’s significantly larger number of parameters (27B compared to Krikri’s 8B) as well as its more modern architecture.

Specifically, the most significant difference is observed in the subject of the Greek Language; Gemma scores 71.9%2 accuracy and 40.08% ChrF, while krikri falls behind, scoring 46.06 and 21.03, respectively. With respect to Mathematics, both models have a lower performance, which is reasonable given the inherent complexity of the subject and the challenges of processing mathematical reasoning in Greek. Notably, in terms of the ChrF metric, both models score comparatively similarly (Gemma-26.28 vs Krikri-24.26), revealing krikri’s potential attempt to mimic the correct answer’s structure, despite failing to provide the correct final output (27.93%Accuracy).

Concerning Physics, which consists only of open-ended tasks, Gemma outperforms Krikri by approximately 4.5 points (24.02 vs 19.48). On the other hand, the subject of Religious Studies is comprised solely of closed questions; it is observed that both models have higher scores compared to the other three subjects (77.78% for Gemma and 62.22% for KriKri). This is likely due to the nature of the subject, which requires knowledge recall rather than complex reasoning.

4.2 Impact of Multimodality

Table 2 presents the impact of multimodality on models’ performance. As expected, both models show a significant performance drop when processing multimodal tasks compared to text-only queries. Specifically, Gemma’s accuracy decreases from 63.78% to 44.79%, while Krikri’s falls from 41.50% to 29.17%. This finding highlights the difficult nature of multimodal reasoning, requiring visual and textual information integration.

In closed-ended tasks, Gemma maintains a clear lead in both multimodal and non-multimodal ques-

Multimodality	Total Qs	Accuracy (%)		ChrF	
		Gemma-3-27B	KriKri-DPO	Gemma-3-27B	KriKri-DPO
Text-only (No)	1376	63.78	41.50	34.98	22.51
Multimodal (Yes)	270	44.79	29.17	23.74	23.67

Table 2: Performance comparison between text-only and multimodal tasks (RQ2).

Format	Total Qs	Accuracy (%)		ChrF	
		Gemma-3-27B	KriKri-DPO	Gemma-3-27B	KriKri-DPO
Fill in the gaps	65	88.89	77.78	46.99	11.28
Matching	4	75.00	50.00	–	–
Multiple Choice	1268	59.23	37.70	–	–
Open Ended	222	–	–	28.00	25.75
True/False	87	85.06	65.52	–	–

Table 3: Performance comparison across different question formats (RQ3).

tions, demonstrating a superior capability in choosing the correct answer even when visual context is required. However, the difference between Gemma’s and Krikri’s performance is approximately 15 points, revealing a mild deviation between them.

However, a notable observation is detected in open-ended multimodal tasks, where the performance of the two models converges. While Gemma outperforms Krikri in non-multimodal open and closed tasks (63.78% and 34.98% respectively), the landscape changes in open-ended multimodal questions with both models scoring nearly identically (23.74 vs. 23.67). This suggests that visual complexity makes it difficult for Gemma to produce precise answers; its score drops to the same level as the Krikri model in open-ended tasks with visual components. Therefore, even though Gemma selects the correct answer in terms of Accuracy, it struggles to unfold the reasoning derived from the visual input.

4.3 Performance Across Exercise Types

Table 3 presents model performance by exercise type. Multiple-choice questions constitute the majority of the exam dataset (1,268 questions), in which Gemma demonstrates a substantial performance of 59.23% accuracy compared to Krikri’s 37.70%.

Both models achieve the highest accuracy scores

in Fill-in-the-gaps (88.89% and 77.78%) and True/False tasks (85.06% and 65.52%). The high performance in True/False questions is explained on the grounds of the binary nature of the task (50% random chance baseline). However, the performance gap ranges between 10% and 20%, with Gemma remaining superior in reasoning tasks.

Another discrepancy is observed in the Fill-in-the-gaps exercise type regarding open-ended tasks. While Gemma scores a high ChrF metric of 46.99, Krikri drops significantly to 11.28. This suggests that while Krikri is capable of selecting the correct option (77.78%), it struggles when generating the missing text directly due to difficult formatting or hallucinations.

Finally, the two models perform similarly in Open-Ended tasks in terms of textual similarity (Gemma-28 vs. Krikri-25.75); this reinforces the observation from RQ1 that Krikri performs efficiently when producing structurally relevant responses in Greek, even if accuracy in complex reasoning and processing tasks remains lower.

5 Discussion

In our comparative analysis, we investigated the performance of Gemma and Krikri on the GR-ProtipaExams dataset, which consists of four subjects, multiple exercise types (closed and open-ended), along with multimodal context. Our findings reveal that Gemma systematically outper-

formed Krikri across all subjects and task types.

Their overall performance gap averaged around 20%, which can be explained on the grounds that Gemma possesses a significantly larger number of parameters (27B) compared to the Krikri model (8B), alongside a more modern architecture. Despite the fact that Krikri is a Greek finetuned model, it appears incapable of bridging the reasoning gap imposed by the difference in model scale. However, their performance seems to be similar in the subject of Religious Studies (77.78% vs 62.22%), likely due to the nature of the multiple-choice tasks in this domain, which require knowledge recall rather than complex reasoning and processing.

Moreover, in the subject of Mathematics, both models display lower scores in open-ended tasks (Gemma 26.28 vs KriKri 24.26 ChrF), with the results being remarkably close. This finding suggests that Krikri has acquired the logical structure of mathematical problems in Greek (given the similarity in ChrF scores), but at the same time, it fails to provide the correct final solution, as evidenced by its low Accuracy (27.93%) in closed-ended tasks.

It is also observed that when visual information is introduced, the performance of both models drastically drops; Gemma’s accuracy falls from 64% to 45% and KriKri’s from 42% to 30%. With respect to open-ended multimodal questions, a notable convergence is observed, as both models achieve nearly identical ChrF scores (23.74 vs 23.67). This finding suggests that even though Gemma maintains its superiority in selecting the correct answers, the visual complexity imposes a difficulty in generating high-quality textual representation, dropping its performance to the baseline of Krikri model.

Finally, both models perform systematically better and more consistently in closed-ended questions compared to open-ended ones. Specifically, while in the closed task of Fill-in-the-gaps, Krikri achieves a significantly higher score compared to its performance in other exercise types (77.78%), it struggles with the ChrF metric (11.28%), which involves filling in the missing text directly. On the contrary, Gemma is more accurate and precise in terms of providing contextually and factually correct answers (46.99%).

6 Conclusions

In this study, we evaluated the performance of two LLMs, the general-purpose Gemma-3-27B and the Greek-focused KriKri-DPO-Context, on

GR-ProtipaExams, a comprehensive multimodal dataset comprising 1,646 questions from Greek Model and Experimental secondary and high school entrance exams across four subjects: Modern Greek, Mathematics, Physics, and Religious Studies.

Our comparative analysis indicates that Gemma systematically outperformed Krikri across all domains; their performance gap becomes evident in complex subjects such as Modern Greek and Mathematics, ranging from 15 to 25%, while it narrows down to around 10% in Physics and Religious Studies, which rely more on knowledge recall.

In terms of multimodality, the introduction of visual elements proved to be a significant bottleneck, causing a drastic performance drop in closed-ended tasks for both models. Furthermore, our analysis suggests a notable convergence in open-ended multimodal generation; while Gemma remains superior in achieving higher scores in closed-ended multimodal tasks, its ability to generate high-quality and precise textual representations drops in the presence of visual elements. Therefore, its performance is similar to Krikri’s in the same multimodal tasks.

Overall, our findings highlight that while language-specific fine-tuning yields structural consistency in open-ended generation, it cannot fully compensate for the lack of model scale required for complex reasoning and multimodal integration. Consequently, there remains significant scope for improvement in advancing LLMs’ capabilities in low-resource languages, particularly in combining contextually appropriate text generation with robust logical and visual processing.

7 Resources

We release the GR-ProtipaExams dataset as an open-source resource on Hugging Face. To maintain anonymity during the double-blind review process, the official URL will be made publicly available upon acceptance. Additionally, the accompanying codebase, including evaluation scripts, prerequisites, and management commands to use the given dataset, is available for review in our anonymized GitHub repository at https://anonymous.4open.science/r/protipa_exams_dataset-EC30.

8 Limitations and Future Work

Due to source exam data unavailability, our study is subject to certain limitations; the dataset contains

temporal discontinuities due to the lack of source files or correct solutions from the official portal of the Governing Body of Model and Experimental Schools (years 2014, 2015, and 2018). The dataset also excludes essay-based components, given that no official answers are provided. Consequently, our evaluation does not assess the models’ writing capabilities or their performance stability across every consecutive year.

Second, our comparative analysis focuses on two models of differing scales, with a significant disparity in parameter size (8B vs. 27B). While this setup highlights the trade-offs between specialized fine-tuning and model scale, it does not provide a controlled comparison between models of the same architecture or size. Future research could extend this benchmark to a wider array of Greek-speaking LLMs of varying sizes to isolate the impact of fine-tuning more effectively.

Finally, our study relies on automated metrics (ChrF) for open-ended and multimodal tasks, which serve as a solid comparison baseline but may not fully capture semantic nuances. As observed in the multimodal analysis, overlap-based metrics often fail to distinguish between blind text generation and failed visual reasoning, assigning similar scores to models with different capabilities. Furthermore, strict instruction-following tasks (such as "fill-in-the-gaps") demonstrated that verbose outputs can be heavily penalized by such metrics, even if the core answer is partially correct. This suggests that human evaluation would be a valuable addition for validating reasoning quality and adherence to constraints in future work.

9 Ethical Considerations

The GR-ProtipaExams dataset comprises content derived from the Greek Ministry of Education and the Governing Body of Model and Experimental Schools in Greece, which is publicly available and distributed online. Our use of this data is strictly limited to research purposes, aiming to benchmark and advance the capabilities of Large Language Models in the Greek language. We acknowledge the intellectual property rights of the respective educational bodies and ensure that the dataset distribution aligns with fair use principles for non-commercial, educational research.

A primary motivation for this work is to enrich high-quality, multimodal benchmarks for low-resource languages like Greek. By releasing this

dataset and the evaluation code, we aim to contribute to the broader spread and exploitation of AI technological capabilities, enabling the development of more inclusive models that perform robustly beyond English-centric benchmarks.

Lastly, while we treat this as a zero-shot evaluation, we acknowledge the potential risk of data contamination, as the source exam files are publicly available on the web and may have been included in the pre-training corpora of large-scale models like Gemma.

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